Vladislavs Agarkovs, va22017

RISC-V32I SoC

Motivation:

To create a self-sufficient system with a RISC-V processor and a set of peripherals that includes a display.

Result:

The system is capable of:

- Executing programs written in C
- Driving a monochrome display using double buffering
- Provide peripheral control via memory-mapped I/O

The system operates at a stable 20 MHz frequency and is suitable for experiments and further development.

Solution Features:

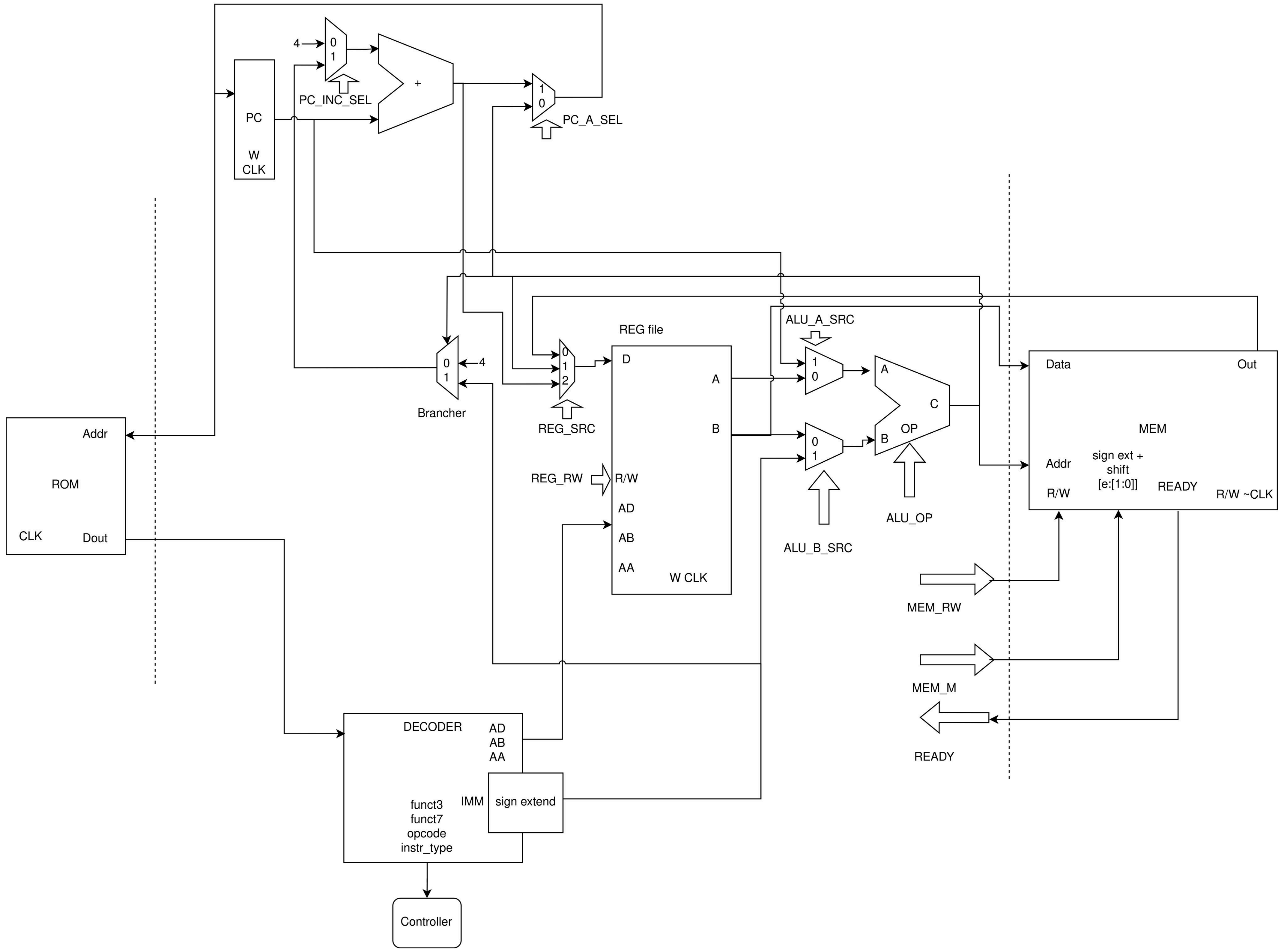
RISC-V 32I single-cycle processor
Processor can read from both RAM and ROM memory
Memory-mapped input/output peripherals

Monochrome display
Display driver with a separate double buffer

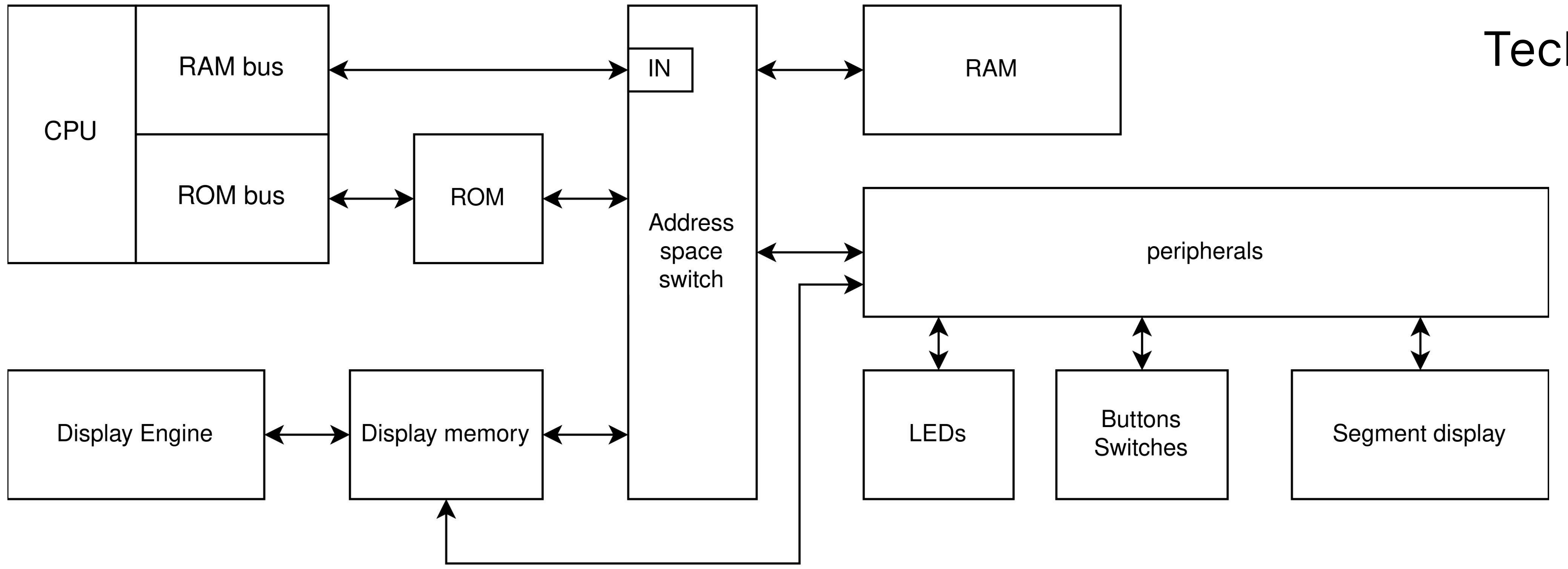
Custom linker script for GCC compiler

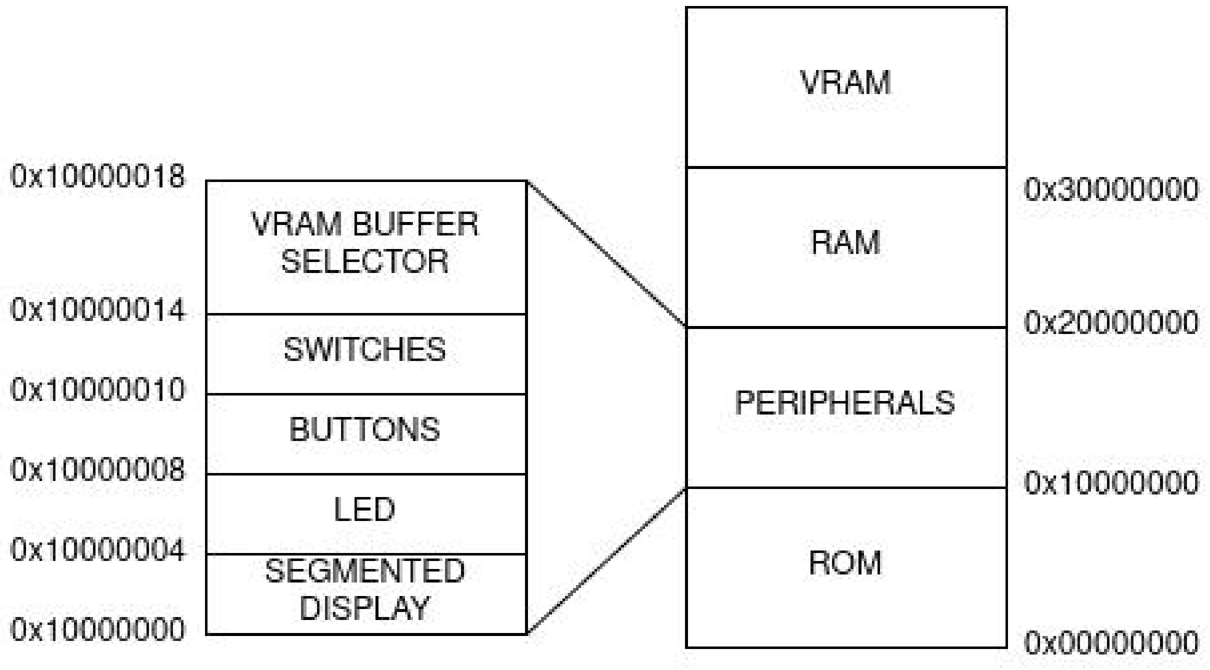
Technical Specification:
65KB ROM
65KB RAM
16KBx2 VRAM
CPU @ 20MHz
LED output
Switches and button inputs
6 seven-segment display outputs
Monochrome screen

RISC-V32I Processor Architecture:



System Architecture:





Conclusions:

Digital design is a complex yet powerful tool for creating hardware. It is easy to make mistakes, finding and fixing bugs is difficult, testing is more than mandatory - testing must be of high quality.